STA 314: Statistical Methods for Machine Learning I

Lecture: K-means clustering

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Unsupervised learning

- Unsupervised learning is the study of learning without labels. What can we do **without labels**?
- How can we even define what learning means without labels?
- In some sense, the ML community does not exactly agree on what it means to do unsupervised learning, but intuitively, unsupervised learning is the task of
  - grouping (clustering)
  - explaining
  - finding structured data

Motivating Examples

Some examples of situations where you'd use unupservised learning

1. You want to understand how a scientific field has changed over time. You want to take a large database of papers and model how the distribution of topics changes from year to year. But what are the topics?

2. You're a biologist studying animal behavior, so you want to infer a high-level description of their behavior from video. You don't know the set of behaviors ahead of time.

3. You want to reduce your energy consumption, so you take a time series of your energy consumption over time, and try to break it down into separate components (refrigerator, washing machine, etc.).

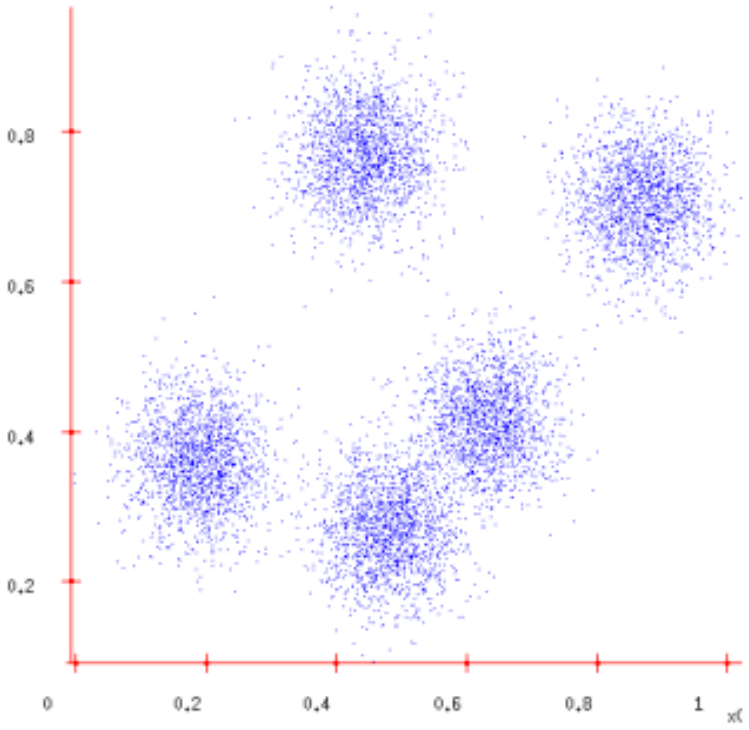
Two major unsupervised learning problems

Common themes: you have some data, and you want to infer some structure underlying the data.

- Clustering
- Low-dimensional representation

Clustering

- Sometimes the data form clusters, where samples within a cluster are similar to each other, and samples in different clusters are dissimilar:

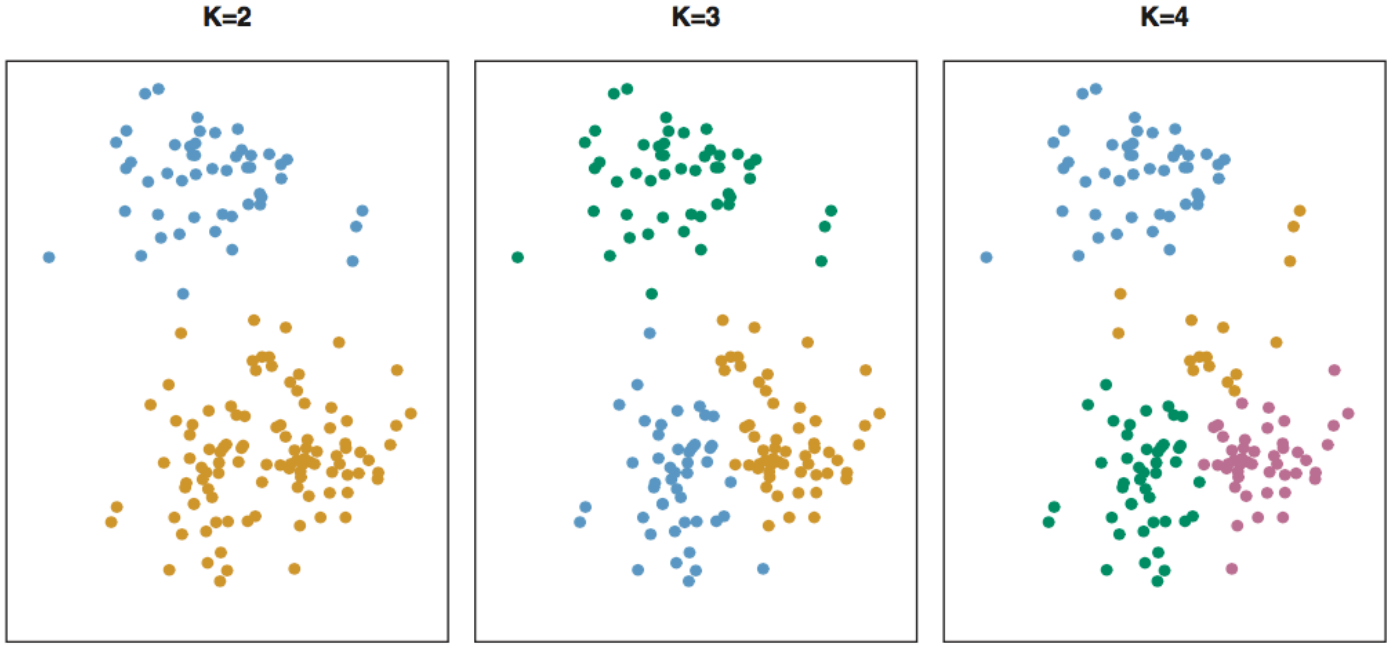


- Such a distribution is **multimodal**, since it has multiple **modes**, or regions of high probability mass.

Clustering

- Grouping data points into clusters, **with no observed labels**, is called **clustering**. E.g.
  - Clustering machine learning papers based on topic (deep learning, Bayesian models, reinforcement learning, etc.)
  - But topics are never observed (unsupervised).
- We will study the most popular clustering algorithm: **K-means**.

Example of results from K-means



- A simulated data set with 150 observations in two-dimensional space. Panels show the results of applying K-means clustering with different values of K.
- The color of each observation indicates the cluster to which it was assigned using the K-means clustering algorithm.
- There is no ordering of the clusters, so the cluster coloring is arbitrary.

K-means Clustering

K-means clustering is a simple approach for partitioning a data set into K **distinct**, **non-overlapping** clusters.

Let $C_1, \ldots, C_K$ denote sets that form a partition of $\{1, \ldots, n\}$:

1. Each observation belongs to at least one of the K clusters.
2. No observation belongs to more than one cluster.

$$C_k \cap C_{k'} = \varnothing, \qquad \text{for all } k \neq k'.$$

Hence every single observation belongs to **one and only one** cluster. The goal is to find such partition $C_1, \ldots, C_K$ for certain purpose.

K-means Clustering

- The idea behind K-means clustering is that a good clustering ensures **the within-cluster variation** as small as possible.
- The within-cluster variation for cluster C_k is a measure $W(C_k)$ on the difference among observations within a cluster.
- We aim to find sets $C_1, \ldots, C_K$ by solving

$$\min_{C_1,\ldots,C_K} \sum_{k=1}^{K} W(C_k).$$

In words, we want to partition the observations into K clusters such that the total within-cluster variation, summed over all K clusters, is as small as possible.

Definition of the within-cluster variation

- It is common to use the Euclidean distance

$$W(C_k) = \frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^{p} \left(x_{ij} - x_{i'j}\right)^2$$
$$= \frac{1}{|C_k|} \sum_{i,i' \in C_k} \|\mathbf{x}_i - \mathbf{x}_{i'}\|_2^2$$

  where $|C_k|$ denotes the number of observations in the kth cluster.

- Let

$$\bar{\mathbf{x}}_k = \frac{1}{|C_k|} \sum_{i \in C_k} \mathbf{x}_i.$$

  Verify that

$$W(C_k) = \frac{1}{|C_k|} \sum_{i \in C_k} \|\mathbf{x}_i - \bar{\mathbf{x}}_k\|_2^2.$$

Target of the K-means algorithm

- Thus, K-means aims to solve

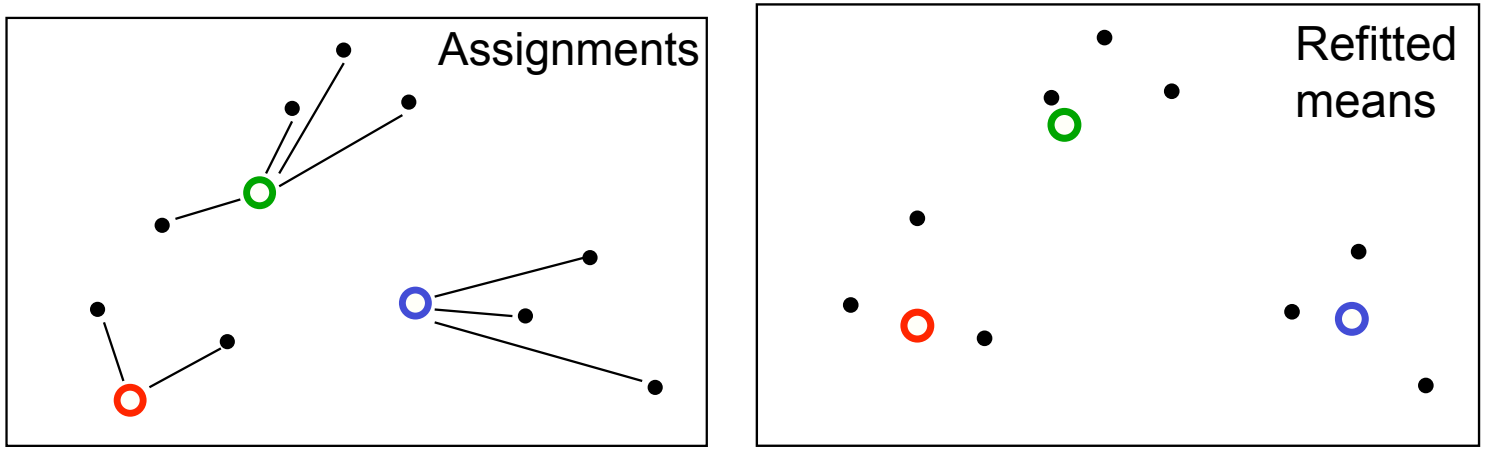
$$\min_{C_1,\ldots,C_K} \sum_{k=1}^{K} W(C_k) \;=\; \min_{C_1,\ldots,C_K} \sum_{k=1}^{K} \frac{1}{|C_k|} \sum_{i \in C_k} \|\mathbf{x}_i - \bar{\mathbf{x}}_k\|_2^2.$$

- This is, however, a very difficult problem to solve exactly. There are almost K^n ways to partition n observations into K clusters.

A pratical alternating algorithm

High level overview of algorithm:

- **Initialization**: randomly initialize cluster centers
- The algorithm iteratively alternates between two steps:
  - **Assignment step**: Assign each data point to the closest cluster
  - **Re-center step**: Move each cluster center to the mean of the data assigned to it



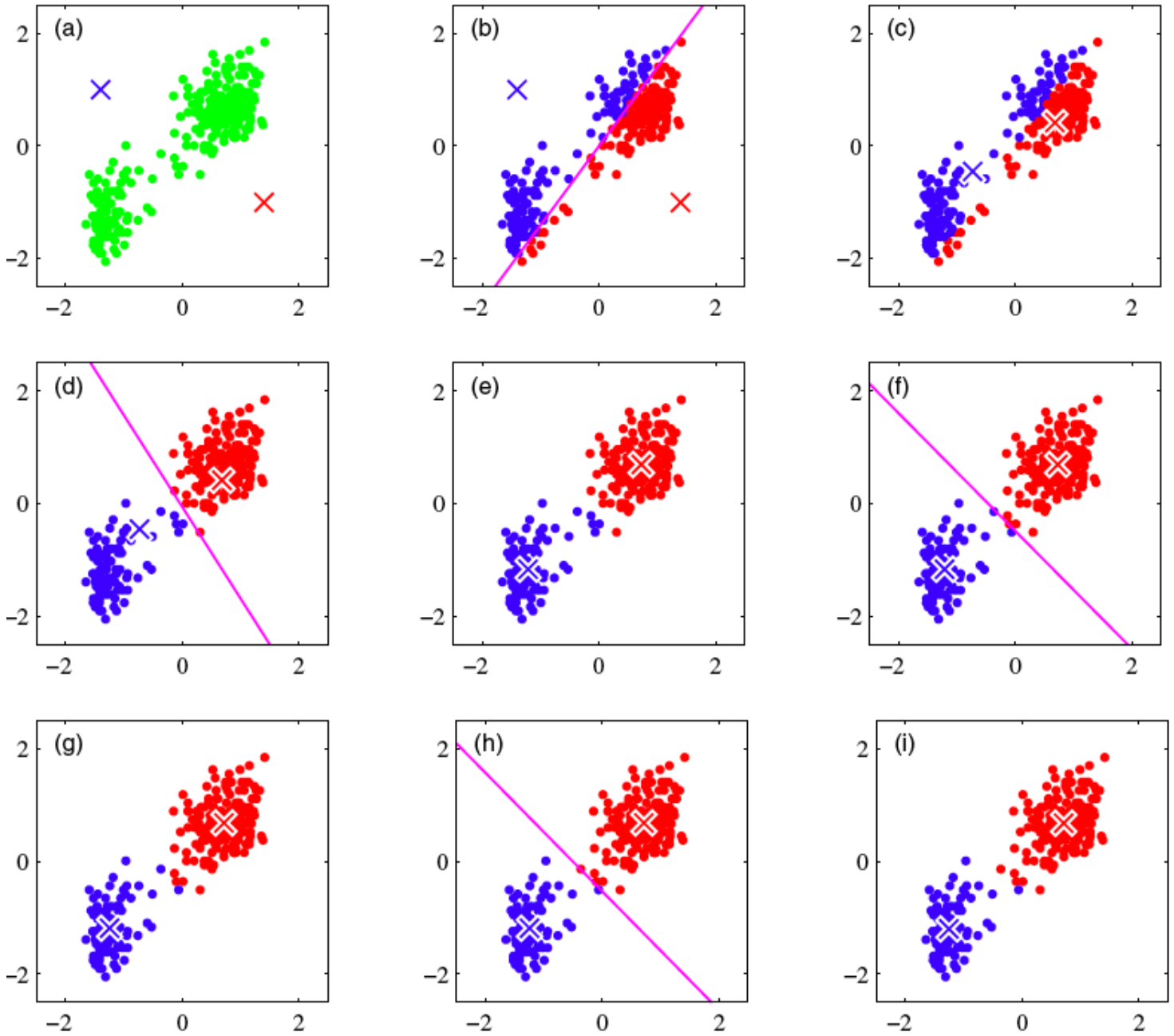


Figure from Bishop

Animation of the algorithm: `http://shabal.in/visuals/kmeans/5.html`.

K-means algorithm

Algorithm 10.1 *K-Means Clustering*

1. Randomly assign a number, from 1 to K, to each of the observations. These serve as initial cluster assignments for the observations.

2. Iterate until the cluster assignments stop changing:

   (a) For each of the K clusters, compute the cluster *centroid*. The kth cluster centroid is the vector of the p feature means for the observations in the kth cluster.

   (b) Assign each observation to the cluster whose centroid is closest (where *closest* is defined using Euclidean distance).

Why K-means Converges

- K-means algorithm reduces the cost at each iteration.

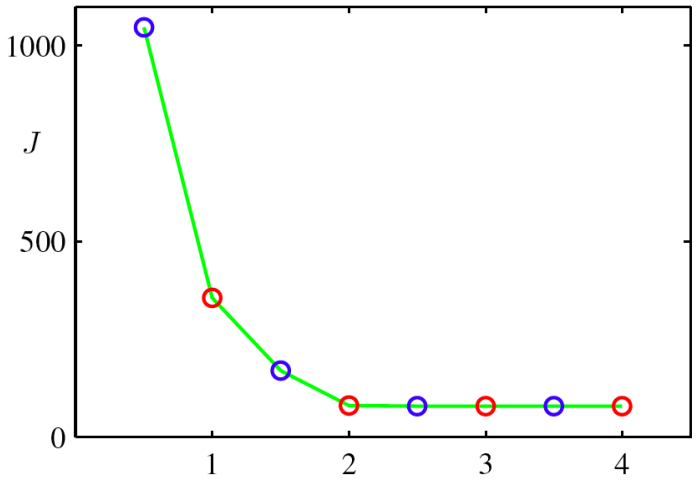
$$\min_{C_1,\ldots,C_K} \sum_{k=1}^{K} \frac{1}{|C_k|} \sum_{i \in C_k} \|\mathbf{x}_i - \bar{\mathbf{x}}_k\|_2^2.$$

  - Assignment step: fixing the centers, re-assignment will decrease the total within-cluster variation.
  - Re-centering step: fixing the assignments, re-centering the data within clusters will reduce the total within-cluster variation.

- Stopping criterion for convergence: when the assignments do not change in the assignment step, we have converged (to at least a local minimum).

Convergence of the K-means algorithm

- Convergence will always happen after a finite number of iterations, since the number of possible cluster assignments is finite

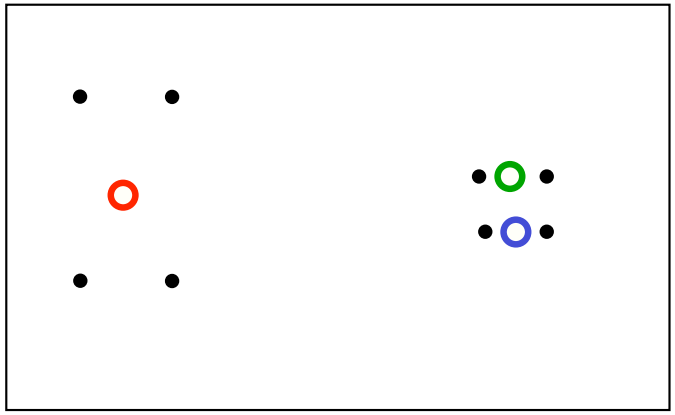


- K-means cost function after each assignment step (blue) and refitting step (red). The algorithm has converged after the third refitting step.

Local Minima

- The cost function is non-convex (so convergence is not equivalent to the global minimum)
- There is nothing to prevent K-means getting stuck at local minima.
- Possible remedy: could try many random starting points

A bad local optimum



K-means for Vector Quantization

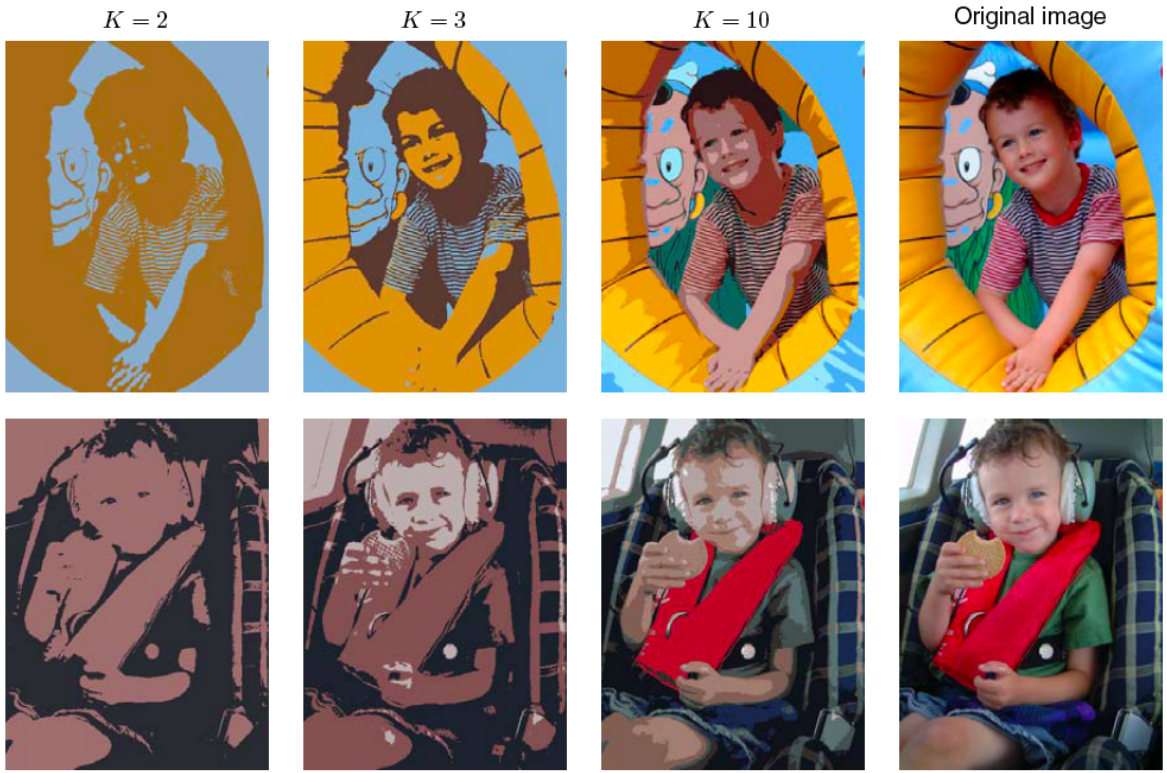
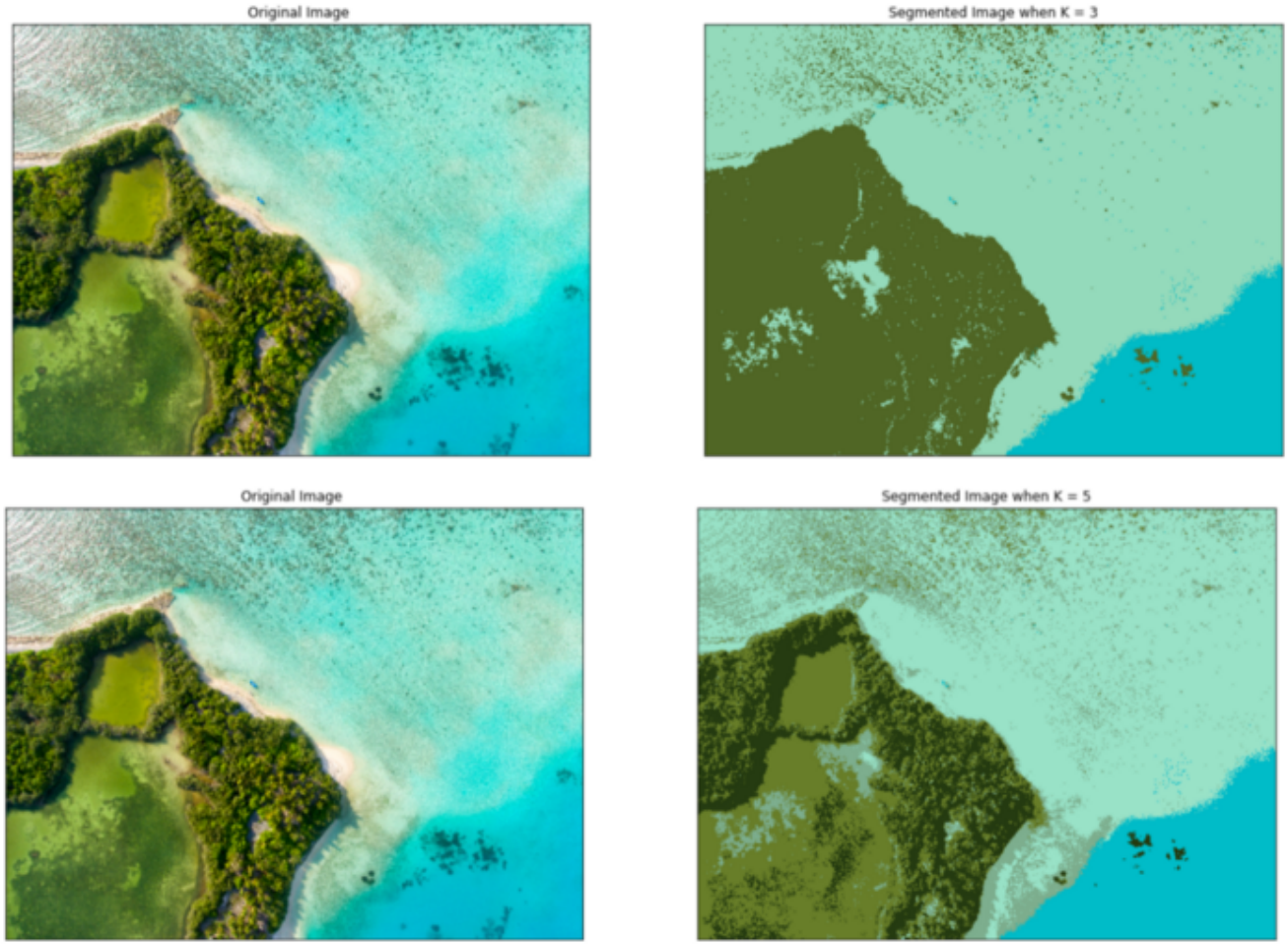


Figure from Bishop

- Given image, construct "dataset" of pixels represented by their RGB pixel intensities
- Run K-means, replace each pixel by its cluster center

K-means for Image Segmentation



- Given image, construct "dataset" of pixels, represented by their HSV pixel intensities
- Run K-means to get superpixels

Extensions

- **Non-exhaustive** clustering. Allow some of the data points not to belong to any cluster.
- **Overlapping** clustering. Allow some of the data points to belong to more than one clusters.
  - Soft K-means
- Clustering **features** rather than data points. Previously, we consider the clustering for data points.